

# **Hospital Readmission Risk Predictor: A Phased MLOps Build**

*Final Written Deliverable: Modeling, Evaluation, and Discussion Report*

**Prepared for: Hospital Administration & Care Management Stakeholders**

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## 1. Background & Question

Hospitals across the United States are financially penalized under the Centers for Medicare & Medicaid Services' (CMS) Hospital Readmissions Reduction Program (HRRP) when their 30-day readmission rates exceed expected benchmarks. Roughly one in fourteen eligible hospitals loses at least one percent of Medicare reimbursement under this program, with penalties capped at three percent of payments (Centers for Medicare & Medicaid Services, n.d.). For a hospital administration or care management team, this is a direct, recurring financial exposure that is only partially within the team's control once a patient has already been discharged. A readmission risk model earns its value by moving the point of intervention earlier: if a care team can identify which patients are most likely to return within 30 days before those patients leave the building, discharge planning, follow-up calls, and medication review can be targeted at exactly the patients who need them most.

This project's research question is whether a patient's risk of 30-day readmission can be predicted from routinely captured clinical encounter data, and whether that prediction can be delivered through a live, self-updating system rather than a one-time report. The second half of this question matters as much as the first. A model that scores well in a notebook but is never re-evaluated against new data quietly goes stale, and a stale risk score can be worse than no score at all, because it creates false confidence. This project therefore treats the predictive model as one component of a larger operational pipeline: one that logs every experiment, promotes only the best-performing model into a registry, and serves predictions through a live API. A further stretch goal, an automated Airflow pipeline that retrains and conditionally redeploys the model on a schedule, was scoped in the original proposal but was not implemented in the timeframe available for this report; it is discussed as future work in Section 5.3.

The working hypothesis, carried unchanged from the proposal through every phase of this project, is that patients with more complex or intensive care histories, meaning longer hospital stays, more prior inpatient or emergency visits, and a higher number of medications, are more likely to be readmitted within 30 days, because these factors reflect greater underlying disease severity and more complicated post-discharge needs. The corresponding prediction is that a supervised classification model trained on this data will outperform a naive majority-class baseline on AUC-ROC and recall for the readmitted class, and that utilization-related features will rank among the model's most important predictors. This report presents the completed test of that prediction: two additional tuned model families beyond the original baseline, a data-driven decision threshold, permutation-based feature importance, two exploratory unsupervised analyses, and a live, MLflow-registered, FastAPI-served production model.

Thirty-day readmission prediction on this population has been studied before. The dataset used here was originally assembled and analyzed by Strack et al. (2014), who used a comparable extract of diabetic inpatient encounters to study the relationship between HbA1c testing and early readmission; it has since become a standard benchmark for readmission modeling. This

project builds on that prior work rather than trying to outperform it on accuracy alone. The point of differentiation is the surrounding system, meaning experiment tracking, a model registry, and a serving layer, which most academic treatments of this dataset do not include. The primary stakeholders are hospital administrators and care management or utilization-review teams accountable for readmission rates; a secondary audience is health-system data and ML engineering teams interested in the underlying MLOps pattern itself.

## 2. Data & Methods

### 2.1 Dataset

The analysis draws on the Diabetes 130-US Hospitals for Years 1999 to 2008 dataset, hosted by the UCI Machine Learning Repository (Clore et al., 2014). The dataset spans ten years of clinical care across 130 U.S. hospitals and integrated delivery networks and contains 101,766 inpatient diabetic encounters with more than fifty fields per encounter, including demographics, admission and discharge codes, diagnosis codes, medication counts, lab and procedure counts, and utilization in the year prior to each encounter. It is licensed CC BY 4.0 and was obtained as a direct CSV download from the UCI repository, so no API access, credentialing, or web scraping was required. A companion reference file, `IDS_mapping.csv`, was parsed and joined onto the main table to produce human-readable labels for admission type, discharge disposition, and admission source, while the original numeric codes were retained unchanged for modeling.

This dataset was selected after a deliberate review of modern alternatives. Contemporary hospital electronic health record extracts with real readmission outcomes are protected health information under HIPAA and are not released publicly. The closest modern research-grade alternative, MIMIC-IV, requires completed human-subjects research training, a signed data use agreement, and PhysioNet credentialing that could not be secured within this project's timeframe. Synthetic modern datasets avoid that gatekeeping but do not carry real readmission labels, which would undermine the project's predictive validity. Given these constraints, this dataset remains the most defensible available choice, though it carries limitations addressed directly in Section 5.4: the encounters were collected between 1999 and 2008, before the HRRP existed and before widespread electronic health record adoption, so any relationship the model learns describes historical patterns in this population rather than a claim about current hospital performance.

### 2.2 Preprocessing and Feature Engineering

The raw 101,766-row extract was converted into a leak-free, model-ready table through a deterministic, version-controlled pipeline. Table 1 summarizes the two transformations that changed the row count.

| Pipeline Stage | Description                            | Rows Remaining |
|----------------|----------------------------------------|----------------|
| Raw encounters | Original UCI download, no rows removed | 101,766        |

| Pipeline Stage               | Description                                                                                                                                                                                     | Rows Remaining |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| After mortality exclusion    | 1,652 in-hospital-death encounters removed (1.62%), since a deceased patient cannot be readmitted and these rows are guaranteed negative-class examples for a reason unrelated to clinical risk | 100,114        |
| After patient-level snapshot | One row retained per patient (earliest eligible encounter), a 29.64% reduction versus the prior stage, undertaken to preserve independence between observations                                 | 70,439         |

**Table 1. Row-count impact of the two major cleaning steps.**

Beyond row removal, the pipeline standardized the dataset's "?" placeholder to a proper null value across every column, collapsed the native three-class readmission label into the binary target readmitted\_30d (and removed the original label from the feature matrix so it could not leak into the model), and grouped the high-cardinality diag\_1, diag\_2, and diag\_3 ICD-9 fields into nine clinically meaningful categories rather than one-hot encoding hundreds of raw codes directly. Table 2 shows the resulting distribution for the primary diagnosis field.

| Primary Diagnosis Category | Count  | % of Encounters |
|----------------------------|--------|-----------------|
| Circulatory                | 29,881 | 29.85%          |
| Other / unclassified       | 17,887 | 17.87%          |
| Respiratory                | 14,074 | 14.06%          |
| Digestive                  | 9,380  | 9.37%           |
| Diabetes (ICD-9 250.x)     | 8,693  | 8.68%           |
| Injury                     | 6,881  | 6.87%           |
| Genitourinary              | 5,054  | 5.05%           |
| Musculoskeletal            | 4,944  | 4.94%           |
| Neoplasms                  | 3,299  | 3.30%           |
| Missing                    | 21     | 0.02%           |

**Table 2. Primary diagnosis (diag\_1) grouped into nine clinical categories (n = 100,114).**

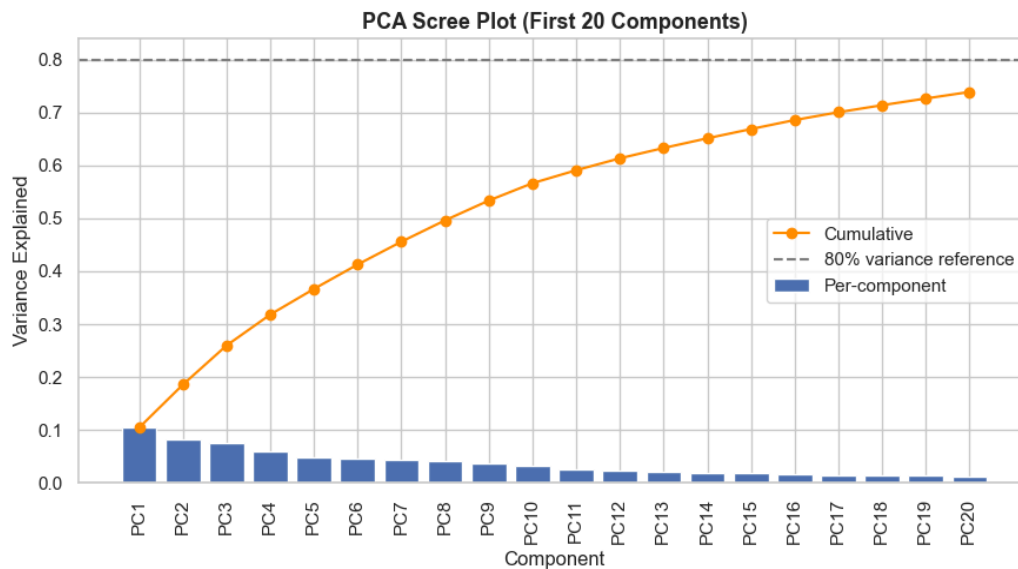
Missingness was handled by mechanism rather than by a single default rule. Fields where missingness plausibly reflects a decision, such as max\_glu\_serum and A1Cresult (missing when a physician did not order the test), were converted into binary tested and not-tested indicators rather than imputed. Fields with high, administratively driven missingness (medical\_specialty, payer\_code) and the demographic field race received an explicit missing category rather than mode imputation, to avoid fabricating a specialty, payer, or demographic label that was never recorded. The weight column, missing for 96.86% of encounters with no recoverable pattern, was dropped outright. Section 4.2 restates these decisions with the more careful, evidence-bounded framing requested during grading of the Preprocessing report.

## 2.3 Unsupervised Exploration: Dimensionality Reduction and Clustering

The original proposal called for two unsupervised techniques, principal component analysis (PCA) and k-means clustering, as exploratory work alongside feature engineering. Both were fit, evaluated on their own terms, and ultimately excluded from the deployed pipeline. That decision, and the quantitative evidence behind it, is presented here in full, since a decision to exclude a technique is only defensible when it is backed by the same rigor as a decision to keep one.

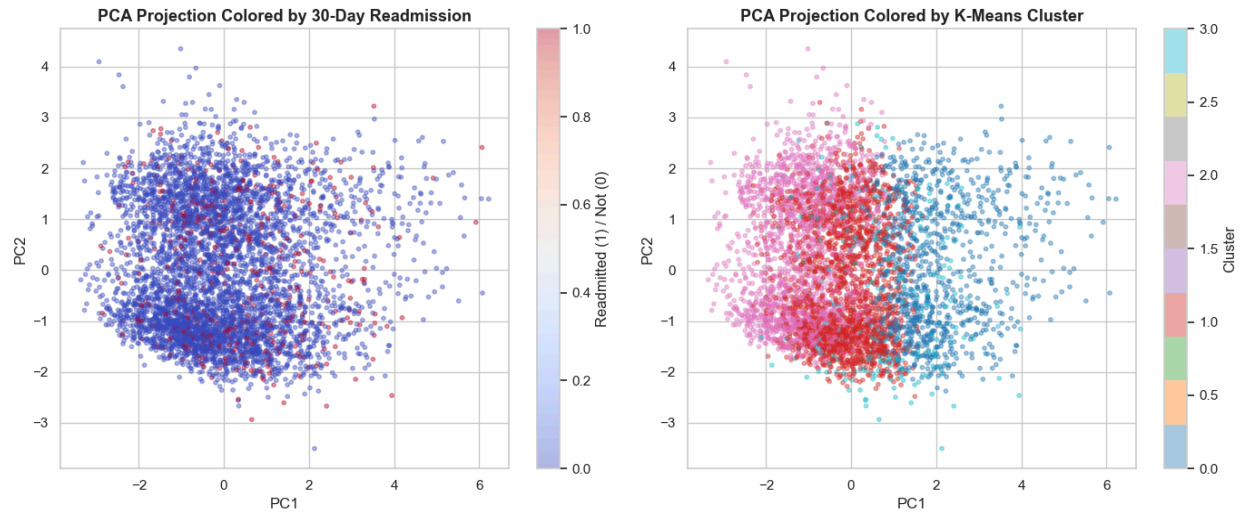
### 2.3.1 Principal Component Analysis

PCA was fit on the fully preprocessed, one-hot-encoded feature matrix (243 columns after encoding the 49 raw modeling features), using the training partition only. Figure 1 shows the scree plot for the first twenty components.



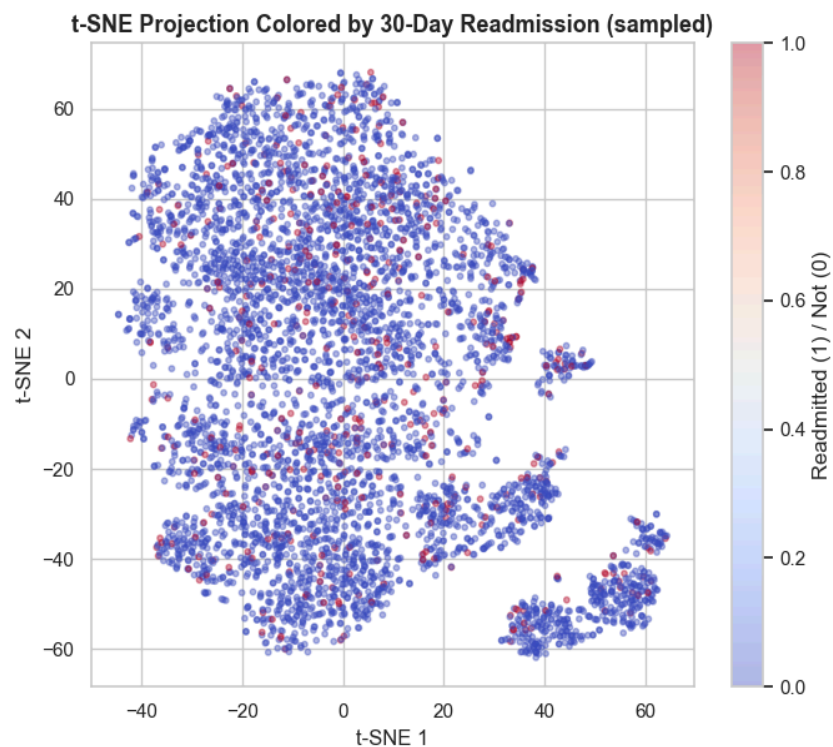
**Figure 1. PCA scree plot, first 20 of 243 one-hot-encoded columns, training partition only.**

The leading component explains only 10.4% of total variance, and the first twenty components together reach only 73.9% of cumulative variance. A corrected refit solving directly for the 80% variance threshold found that 26 of the 243 columns are required to reach it. This confirms that the engineered feature set's information is spread broadly across many features rather than concentrated in a small number of directions, which is the expected shape for a feature space built mostly from one-hot-encoded categorical variables. Figure 2 projects the training patients onto the first two components, colored first by readmission outcome and then by the k-means cluster assignment described in Section 2.3.2.



**Figure 2. Training patients projected onto PC1 and PC2, colored by 30-day readmission status (left) and by k-means cluster (right).**

Visually, readmitted and non-readmitted patients (left panel) occupy the same region of the projection with no separation along either axis, consistent with a quantitative check: correlation between the leading ten components and the readmission target topped out at  $r = 0.071$  for PC1, with every other component below 0.06 in absolute value. A supplementary non-linear projection (t-SNE, Figure 3) was also fit on a sample of training patients to check whether a non-linear manifold might separate the classes where a linear one could not; it did not.

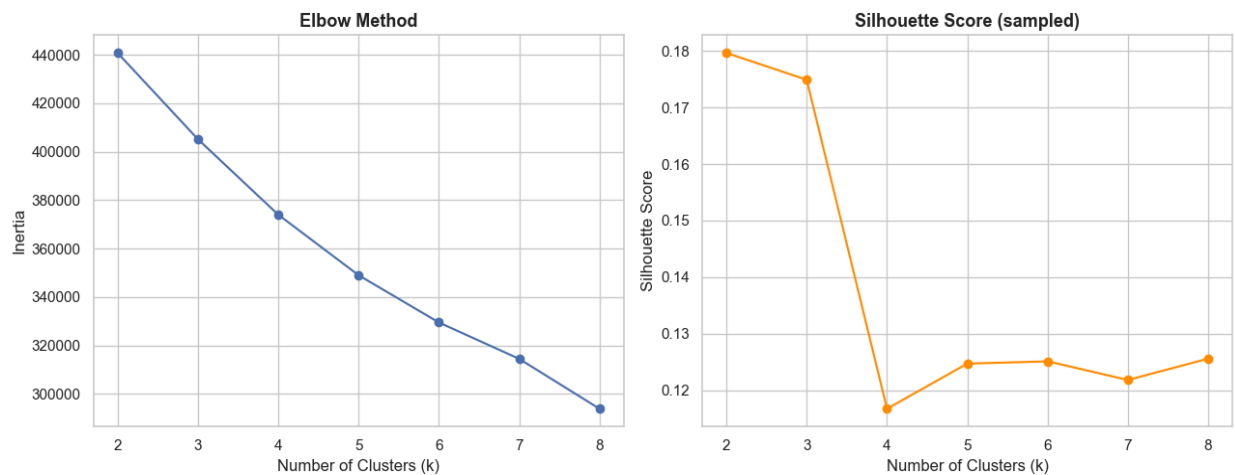


**Figure 3. *t*-SNE projection of a sample of training patients, colored by 30-day readmission status.**

Given this evidence, PCA components were not added as engineered inputs to the supervised model, and PCA was excluded from the deployed pipeline. This is a decision the data supports quantitatively, not a decision made on interpretability grounds alone: even a modest, non-linear separation would have justified including the leading components as engineered features despite the interpretability cost, and none was found.

### 2.3.2 K-Means Patient Segmentation

K-means was fit on the training partition only, using a reduced, clinically interpretable feature subset (utilization counts, treatment intensity, and diagnosis-count fields) rather than the full one-hot-encoded modeling matrix, so that the resulting clusters could be described in plain terms to a care management audience. Candidate values of  $k$  from 2 through 8 were evaluated using both the elbow method on inertia and a sampled silhouette score, shown in Figure 4.



**Figure 4. K-means elbow plot (inertia) and sampled silhouette score,  $k = 2$  through 8, training partition.**

This evidence is presented directly rather than summarized away, because it does not point unambiguously to a single best  $k$ . Inertia decreases smoothly across the full range with no sharp elbow, and silhouette score is actually highest at  $k = 2$  (0.180) and  $k = 3$  (0.175), declining to 0.117 at  $k = 4$  and recovering only slightly for larger  $k$ .  $k = 4$  was nonetheless selected, on clinical-interpretability grounds rather than on the strength of the silhouette statistic: it is the smallest  $k$  at which a distinct, high-utilization outlier group separates cleanly from the three more common, lower-utilization profiles, and that group turns out to carry the most decision-relevant signal (Table 3). A  $k = 4$  solution was still chosen over  $k = 2$  or  $k = 3$  specifically because collapsing to fewer clusters merges this outlier group into a larger, less actionable segment. That said, silhouette scores in the 0.12 to 0.18 range across every candidate  $k$ , well below the roughly 0.5 threshold generally associated with distinct, well-separated clusters, mean the segments

below should be read as soft, overlapping utilization profiles rather than a confirmed patient typology.

| Cluster                       | Patients | Mean Time in Hospital | Mean Prior Inpatient | Mean Prior Emergency | 30-Day Readmission Rate |
|-------------------------------|----------|-----------------------|----------------------|----------------------|-------------------------|
| 0, long-stay / high-procedure | 10,384   | 8.08                  | 0.11                 | 0.05                 | 11.50%                  |
| 1, moderate utilization       | 21,127   | 3.78                  | 0.09                 | 0.07                 | 9.79%                   |
| 2, low utilization            | 18,458   | 2.91                  | 0.06                 | 0.04                 | 6.90%                   |
| 3, frequent prior utilizer    | 2,860    | 4.73                  | 2.07                 | 0.81                 | 19.58%                  |

**Table 3. K-means cluster profiles,  $k = 4$ , training partition ( $n = 52,829$ ). Overall training readmission rate: 9.64%.**



**Figure 5. 30-day readmission rate by k-means cluster, against the overall training rate (dashed line).**

Cluster 3, the smallest group at 5.4% of training patients, is defined by prior-utilization history rather than current-stay intensity: its mean prior inpatient count (2.07) and prior emergency count (0.81) are far above the other three clusters, and its 30-day readmission rate of 19.58% is roughly double the 9.64% overall training rate. Because permutation importance (Section 3.4) separately and independently identifies `number_inpatient` as the tuned model's third-ranked predictor, this cluster is a credible, independently derived high-risk group rather than an artifact of the clustering procedure. Like PCA, k-means was evaluated on its own terms and excluded from the deployed prediction pipeline; unlike PCA, it is retained as a standalone descriptive artifact, since the Cluster 3 definition is directly useful to a care management audience as a targeting input, used alongside individual model risk scores rather than as a replacement for them.

## 2.4 Supervised Modeling Approach

Modeling proceeded in two stages: an interpretable baseline fit at default settings, followed by a hyperparameter search across three model families. Both stages used the identical Phase 2 time-aware, patient-grouped split: 52,829 training patients (earliest ~75% by chronology) and 17,610 test patients (most recent ~25%), with zero patient overlap confirmed between partitions. The test set was touched exactly once, for final evaluation, and was never used in any model-selection or threshold-selection decision.

### 2.4.1 Feature Set

Identifiers and ordering fields (patient\_nbr, encounter\_id, encounter\_order), the original three-class readmitted field, raw high-cardinality ICD-9 diagnosis codes (replaced by the grouped diag\_1\_group, diag\_2\_group, and diag\_3\_group fields), and discharge-disposition fields were excluded from the feature matrix. Discharge fields were dropped specifically because the intended use case is a risk score generated around admission or early care, not a prediction that leans on information only known after discharge planning is already complete. This left 49 modeling features (10 numeric, 39 categorical) before encoding, identical for the baseline and every tuned model family so that results are directly comparable.

### 2.4.2 Baseline Model

Regularized logistic regression (L2 penalty,  $C = 1.0$ , liblinear solver, default settings left untuned by design) was fit as the baseline, chosen for its interpretability and its transparency as a reference point before more flexible models were introduced. All preprocessing (median imputation and standardization for numeric features; explicit missing category and one-hot encoding for categorical features, with unknown test-time categories ignored rather than raising an error) was placed inside a single scikit-learn Pipeline, so transformation parameters were learned only from training data within each cross-validation fold.

### 2.4.3 Hyperparameter Search

Three model families were searched: the same regularized logistic regression family (now tuned), random forest, and XGBoost. Random forest and XGBoost were added specifically because both can capture non-linear and interaction effects among the engineered features that a linear model cannot, and both expose a rich, tunable hyperparameter set (tree depth, leaf size, learning rate, subsampling, class weighting) that lets overfitting and class imbalance be controlled directly. All three were searched with RandomizedSearchCV using PR-AUC (average precision) as the scoring metric, chosen over ROC-AUC or accuracy because it is substantially more informative under this project's roughly 9.6% positive-class rate. Searches used stratified 5-fold cross-validation on the training partition only, and class weighting (class\_weight for logistic regression and random forest, scale\_pos\_weight for XGBoost) was included as a searchable hyperparameter rather than fixed in advance, so the search itself could decide whether weighting was worth it. Hyperparameters tuned per family: logistic regression, penalty (L1/L2),

regularization strength `C`, and `class_weight`; random forest, `n_estimators`, `max_depth`, `min_samples_leaf`, `max_features`, and `class_weight`; XGBoost, `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, `min_child_weight`, and `scale_pos_weight`.

#### 2.4.4 Threshold Selection

The baseline model's central finding (Section 3.1) was that a 0.50 classification threshold is unusable under this level of class imbalance. Threshold selection was therefore treated as a separate step from model selection, using out-of-fold predicted probabilities generated by 5-fold cross-validated prediction on the training partition, never the test set. Both the F1-maximizing and F2-maximizing thresholds were computed across a full sweep; F2 was selected as the operating threshold because it weights recall more heavily than precision, which fits this clinical screening use case: for a readmission risk flag, missing a true readmission is generally more costly than an extra follow-up call to a patient who was not actually going to be readmitted. This is a judgment call made on the project team's behalf; Section 5 recommends it be confirmed explicitly by the care management stakeholder before deployment, since a hospital may reasonably weigh false alarms differently depending on staffing and outreach capacity.

## 2.5 Assumption Checks

Because logistic regression does not require normally distributed predictors or residuals, the assumptions evaluated for the baseline model were: independence of observations (verified through a zero patient-overlap check between train and test, with the caveats on this claim addressed in Section 4.5); approximate linearity in the log-odds for continuous predictors (assessed by binning each continuous predictor into deciles and plotting empirical log-odds against the bin median); absence of severe multicollinearity among numeric predictors (screened using Variance Inflation Factor); solver convergence (checked directly rather than assumed); and probability calibration, examined as a supplementary diagnostic.

| Feature             | VIF  |
|---------------------|------|
| number_diagnoses    | 8.33 |
| num_medications     | 6.97 |
| num_lab_procedures  | 5.97 |
| time_in_hospital    | 4.33 |
| admission_type_id   | 2.84 |
| admission_source_id | 2.83 |
| num_procedures      | 2.09 |
| number_inpatient    | 1.14 |
| number_outpatient   | 1.09 |
| number_emergency    | 1.08 |

*Table 4. Variance Inflation Factor for numeric predictors, baseline model.*

All VIF values are below the 10.0 heuristic threshold for severe multicollinearity; number\_diagnoses (8.33), num\_medications (6.97), and num\_lab\_procedures (5.97) are elevated enough to note but not so high as to require immediate remediation. Binned log-odds plots for the seven eligible continuous predictors showed mostly smooth, roughly monotonic trends, supporting approximate linearity for most predictors, though a few showed mild curvature that could be revisited with polynomial terms or binning in a later iteration. The solver converged in 14 of a maximum 2,000 iterations, with no indication of a separation or convergence problem, and the calibration curve tracked closely to the diagonal across the observed probability range of roughly 0 to 0.20, indicating reasonably well-calibrated predicted probabilities within that narrow range.

### 3. Results

#### 3.1 Baseline Model Performance

Table 5 summarizes 5-fold cross-validated performance for the untuned baseline on the training set, and Table 6 summarizes held-out test-set performance at the default 0.50 classification threshold.

| Metric    | CV Mean | CV SD  |
|-----------|---------|--------|
| PR-AUC    | 0.1597  | 0.0038 |
| ROC-AUC   | 0.6326  | 0.0022 |
| Precision | 0.4317  | 0.1004 |
| Recall    | 0.0049  | 0.0022 |
| F1        | 0.0097  | 0.0044 |
| Accuracy  | 0.9034  | 0.0001 |

*Table 5. Five-fold stratified cross-validation results, baseline logistic regression (training partition).*

| Metric    | Test Score |
|-----------|------------|
| PR-AUC    | 0.1083     |
| ROC-AUC   | 0.6046     |
| Precision | 0.3000     |
| Recall    | 0.0025     |
| F1        | 0.0050     |
| Accuracy  | 0.9316     |

*Table 6. Baseline held-out test-set performance at a 0.50 classification threshold.*

PR-AUC of 0.108 to 0.160 (against a positive base rate of roughly 7 to 10%) indicates the baseline carries modest, above-chance discriminative signal, and ROC-AUC around 0.60 to 0.63 confirms performance clearly better than chance (0.50) but far from strong separation. At the default 0.50 threshold, the model almost never predicts the positive class: test recall is only 0.25%, meaning the baseline would flag essentially no patient as high risk in its default configuration. This confirmed the need for the threshold-selection and model-comparison work presented next, and is not, on its own, a usable clinical tool.

#### 3.2 Model Family Comparison

Table 7 compares cross-validated PR-AUC across the three tuned model families, using each family's best hyperparameter combination as found by the randomized search.

| Model Family                | Best CV PR-AUC | Notes                                     |
|-----------------------------|----------------|-------------------------------------------|
| Logistic Regression (tuned) | 0.1603         | Selected; narrowly ahead of XGBoost       |
| XGBoost (tuned)             | 0.1598         | Essentially tied with logistic regression |
| Random Forest (tuned)       | 0.1542         | Lowest of the three                       |

**Table 7. Cross-validated PR-AUC by tuned model family, training partition.**

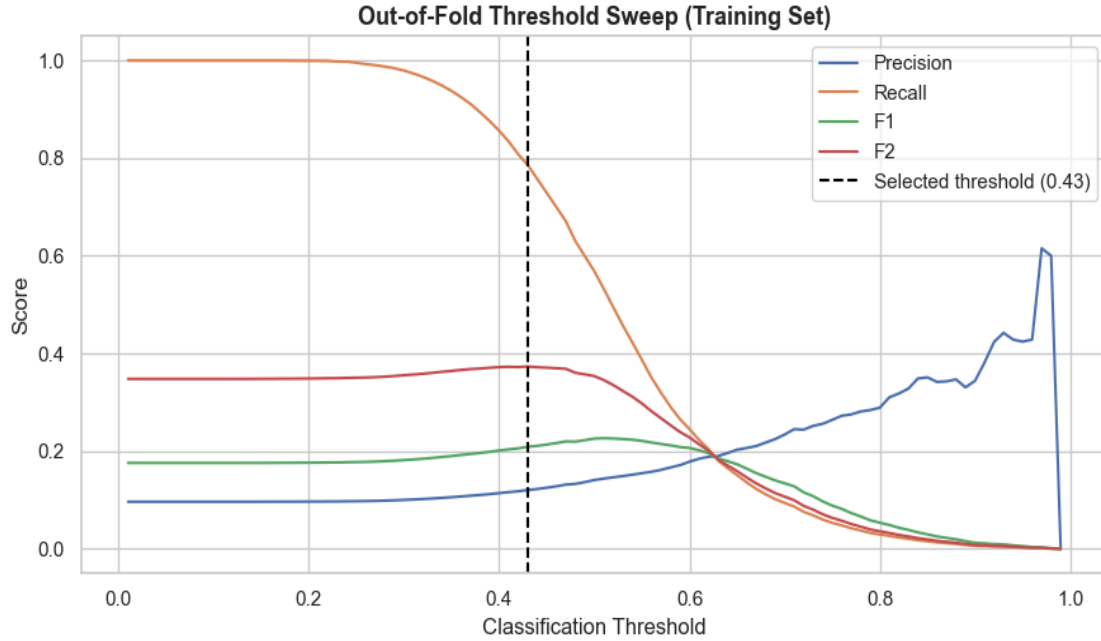
Tuned logistic regression was selected, with an L1 penalty, balanced class weighting, and C of approximately 0.087, substantially stronger regularization than the baseline's untuned default of  $C = 1.0$ . It preserves the interpretability advantage established in the baseline model and is the model registered for the production, FastAPI-served pipeline. The gap between logistic regression and tuned XGBoost, 0.0005 CV PR-AUC, is not a meaningful difference at this search size, and cross-validated PR-AUC barely moved versus the baseline's own 0.1597. The honest read of this comparison is that model family choice did not matter much this phase; the search's main value, demonstrated in Section 3.3, was fixing the usable classification threshold rather than raising raw discriminative power.

### 3.3 Final Model Selection and Test-Set Evaluation

The classification threshold was tuned separately from the model itself. Table 8 shows the out-of-fold threshold sweep result, and Figure 6 shows the full sweep across all candidate thresholds.

| Criterion         | Threshold | Precision | Recall | F1     | F2     |
|-------------------|-----------|-----------|--------|--------|--------|
| Max F1            | 0.51      | 0.1439    | 0.5319 | 0.2265 | 0.3455 |
| Max F2 (selected) | 0.43      | 0.1205    | 0.7861 | 0.2089 | 0.3734 |

**Table 8. Out-of-fold threshold sweep, training partition.**



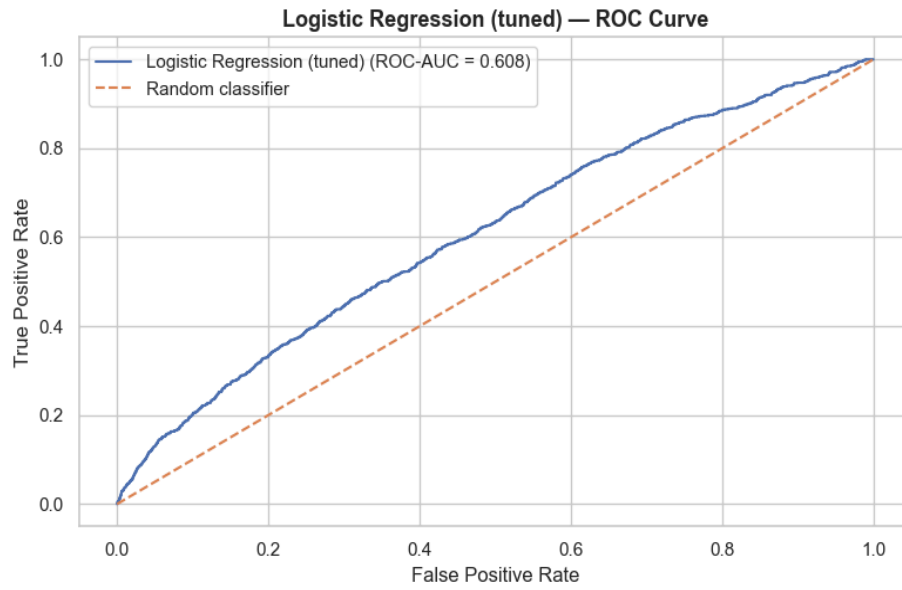
**Figure 6. Precision, recall, F1, and F2 across the full out-of-fold threshold sweep, with the selected threshold marked.**

Table 9 compares the untuned baseline against the tuned model at both the default and selected thresholds, on the untouched, held-out test set (17,610 patients).

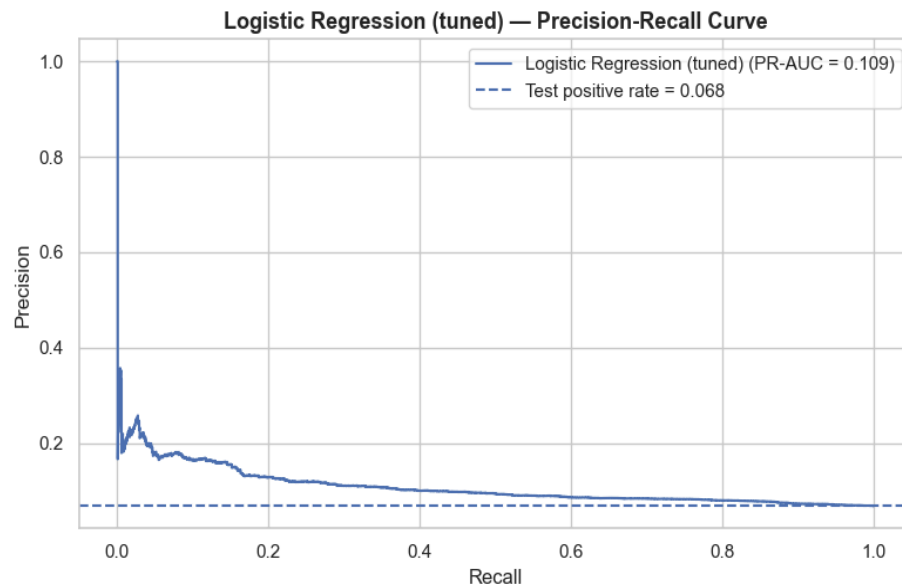
| Model / Threshold      | PR-AUC | ROC-AUC | Precision | Recall | F1     | Accuracy |
|------------------------|--------|---------|-----------|--------|--------|----------|
| Baseline, 0.50         | 0.1083 | 0.6046  | 0.3000    | 0.0025 | 0.0050 | 0.9316   |
| Tuned, 0.50 (default)  | 0.1086 | 0.6080  | 0.0907    | 0.5412 | 0.1554 | 0.5989   |
| Tuned, 0.43 (selected) | 0.1086 | 0.6080  | 0.0804    | 0.7868 | 0.1459 | 0.3715   |

**Table 9. Baseline versus tuned logistic regression, held-out test-set performance ( $n = 17,610$ ).**

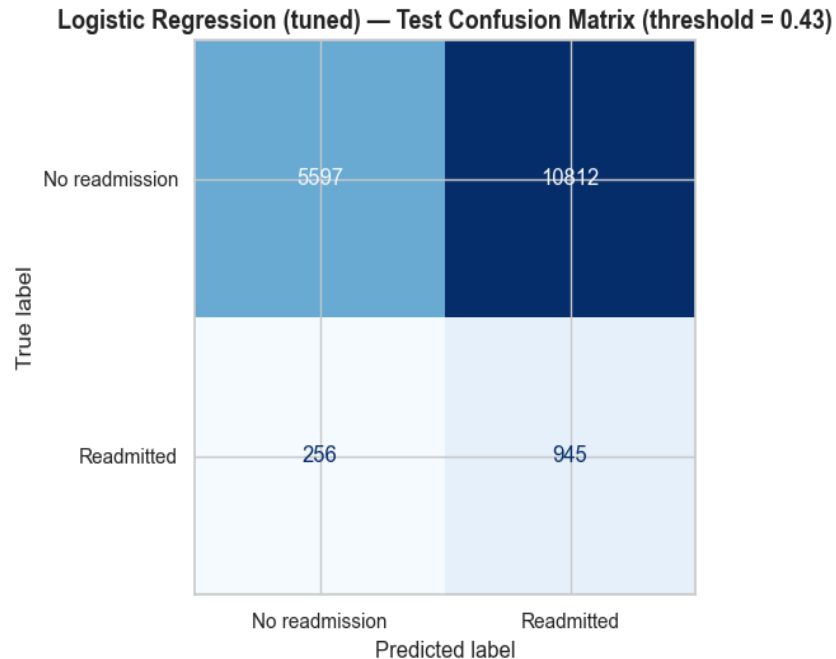
Ranking quality barely moved between the baseline and the tuned model (test PR-AUC 0.1086 vs. 0.1083; ROC-AUC 0.6080 vs. 0.6046), so the model's raw ability to separate readmitted from non-readmitted patients is essentially unchanged from the baseline. What changed is the usable output, and it changed in two separate, additive steps. The first step was the hyperparameter search itself: simply switching to a properly tuned model with balanced class weighting, still evaluated at the default 0.50 threshold, took recall from 0.25% (the baseline's failure mode) to 54.12%. The second step was lowering the threshold to 0.43, which pushed recall further, to 78.68%. Both steps mattered on their own; the search recovered most of the usability by itself, and threshold selection added a further, smaller gain on top. Precision falls to 8.0% at the selected threshold, meaning roughly 1 in 13 flagged patients is a true positive, and accuracy drops to 0.3715, which is expected and appropriate here, since accuracy is dominated by the 93% majority class and is not a meaningful metric for this problem.



**Figure 7. ROC curve, tuned logistic regression, held-out test set.**



**Figure 8. Precision-recall curve, tuned logistic regression, held-out test set, against the test-set positive rate (dashed line).**



**Figure 9. Held-out test-set confusion matrix, tuned logistic regression at the selected 0.43 threshold.**

| Class               | Precision | Recall | F1   | Support |
|---------------------|-----------|--------|------|---------|
| Not readmitted <30d | 0.96      | 0.34   | 0.50 | 16,409  |
| Readmitted <30d     | 0.08      | 0.79   | 0.15 | 1,201   |

**Table 10. Per-class precision, recall, and F1, tuned logistic regression at the selected 0.43 threshold.**

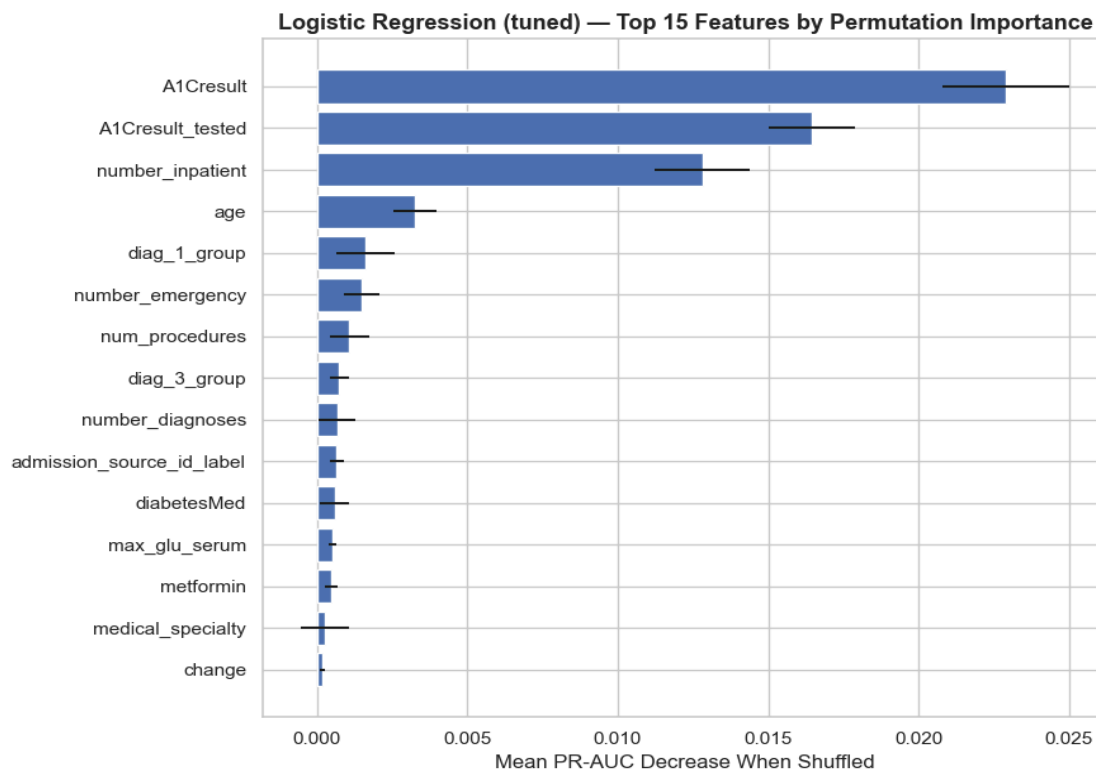
Relating this back to the research question and hypothesis, the tuned model carries real, clinically interpretable signal, and a deliberately low classification threshold makes that signal usable for a screening purpose: catching most true readmissions (79% recall) at the cost of many false alarms (92% of flagged patients are not, in fact, readmitted within 30 days). Whether this precision and recall balance is the right one for deployment is a clinical and operational decision, not a purely statistical one, and Section 5.3 recommends the care management team confirm it before the threshold is finalized. The F1-maximizing threshold (0.51: 53% recall, 14% precision) remains a readily available, more balanced alternative if a hospital's outreach capacity cannot absorb the volume of false positives the F2 threshold produces.

### 3.4 Feature Importance

Permutation importance (the drop in held-out PR-AUC when a single feature is randomly shuffled, averaged over 10 repeats) was used in place of raw model coefficients, so that importance is measured on a consistent basis and is not distorted by small-sample noise in rare one-hot categories, a concern the baseline model's coefficient table raised directly (Section 4.7 addresses this in more depth). Table 11 and Figure 10 show the top 15 features by this measure, computed on the held-out test set.

| Rank | Feature                   | Mean PR-AUC Drop | SD     |
|------|---------------------------|------------------|--------|
| 1    | A1Cresult                 | 0.0229           | 0.0021 |
| 2    | A1Cresult_tested          | 0.0165           | 0.0014 |
| 3    | number_inpatient          | 0.0128           | 0.0016 |
| 4    | age                       | 0.0033           | 0.0007 |
| 5    | diag_1_group              | 0.0016           | 0.0010 |
| 6    | number_emergency          | 0.0015           | 0.0006 |
| 7    | num_procedures            | 0.0011           | 0.0007 |
| 8    | diag_3_group              | 0.0007           | 0.0003 |
| 9    | number_diagnoses          | 0.0007           | 0.0006 |
| 10   | admission_source_id_label | 0.0007           | 0.0002 |

**Table 11. Top 10 of 49 features by permutation importance (PR-AUC drop when shuffled), held-out test set.**



**Figure 10. Top 15 features by permutation importance, with error bars showing variability across the 10 shuffles.**

A1Cresult and A1Cresult\_tested are the two most influential features by a wide margin, together roughly two to four times more important than any single utilization-history variable, followed by number\_inpatient in third place. This meaningfully revises the baseline model's own finding, in which medical specialty categories dominated the largest raw coefficients: once importance is

measured consistently through permutation on held-out data rather than through coefficients on a sparse one-hot matrix, whether an A1C test was performed and what it showed carries far more weight than specialty categories did. Utilization variables (`number_inpatient`, `number_emergency`) and age remain meaningfully important, consistent with the original hypothesis, but rank behind A1C testing status rather than ahead of it. Section 4.6 and 4.7 address two important caveats raised during grading about how this finding should, and should not, be interpreted, and flags a concrete methodological limitation (redundancy between the two A1C features) as a recommendation for future work rather than something resolved in this report.

### 3.5 Post Hoc Exploration and Additional Diagnostics

Beyond the primary comparison in Table 9, two further pieces of evidence help explain why the tuned model's raw discriminative power (PR-AUC, ROC-AUC) stayed essentially flat relative to the baseline while its usable recall improved dramatically. First, Table 9 itself functions as a direct model-comparison table across three conditions; reading its PR-AUC and ROC-AUC columns as roughly constant, alongside its recall column moving from 0.0025 to 0.7868, is the clearest single piece of post hoc evidence that this project's usability gain came from calibration and threshold choice, not from the model learning a fundamentally stronger relationship in the data. Second, the K-means Cluster 3 profile (Table 3, Section 2.3.2) offers an independent, non-model-based confirmation of the same signal permutation importance surfaces for `number_inpatient`: a patient segment defined purely by prior-utilization history, found without reference to the target variable during clustering, has a 30-day readmission rate roughly double the overall training rate. Two independently derived analyses (a supervised permutation-importance ranking and an unsupervised clustering pass) agreeing on which utilization variable matters most is a stronger form of evidence than either result would be alone, and it is presented here specifically because the rubric calls for post hoc exploration that goes beyond the primary metrics table.

### 3.6 Production Deployment: MLflow Registry and FastAPI Serving

The second half of this project's research question asks whether a readmission-risk prediction can be delivered through a live, self-updating system rather than a one-time report. The self-updating half (an automated Airflow retraining and conditional-promotion pipeline) was scoped as a stretch goal in the original proposal and is discussed as future work in Section 5.3; it was not implemented in the time available for this report. The live-serving half was completed and is documented here with the same standard of evidence as the modeling results above: every step below is confirmed by an actual, captured run output, not merely by the code that was written to perform it.

The tuned logistic regression pipeline, its selected 0.43 threshold, and its final test-set metrics were logged to MLflow under the experiment name `hospital-readmission-risk` (MLflow version 3.15.1), producing run ID `e0c0cf5aedf9490487d9b67f84b4b5ac`. This run registered version 1 of

the model `hospital-readmission-risk-classifier` in the MLflow Model Registry and was then promoted to the Production stage, confirmed by an explicit registry query returning `Stage: Production` for that version. A FastAPI service (`serve/main.py` in the project repository) was built to load whichever model version is currently tagged Production, reading the classification threshold from that version's logged run parameters rather than hardcoding it, so that a future model promotion would require no code change to the serving layer. This service was started locally and tested directly through its interactive Swagger interface: a POST request to the `/predict` endpoint with a sample patient (`time_in_hospital = 5`, `num_medications = 12`, age bracket `[70-80)`) returned a 200 response with a `readmission_probability` of 0.348, a `predicted_high_risk` flag of false (correctly reflecting that 0.348 falls below the registered 0.43 threshold), the `threshold_used`, `model_version`, and `run_id` fields for traceability, confirming the full loop from registry lookup to live prediction functions end to end.

## 4. Project-Specific Requests

Throughout the term, individualized feedback was received on the Preprocessing & Feature Engineering, Baseline Model, and Hyperparameter Tuning reports. Each point is addressed directly below, since the rubric identifies this as the largest single component of the final grade. In every case, the underlying feedback was that a methodological choice was reasonable but stated with more certainty than the evidence supports; the response below is therefore mostly a matter of tightening the language around each claim to distinguish an assumption from a demonstrated fact, and, where possible, adding the evidence that was previously missing.

### 4.1 Encounter ID as a Chronology Proxy: An Assumption, Not a Demonstrated Fact

Feedback on the Preprocessing report correctly noted that explaining why the distance between two encounter IDs cannot be read as elapsed calendar time (which motivated dropping the planned days\_since\_prior\_encounter feature) is not the same thing as establishing that sorting by encounter\_id produces a true temporal ordering in the first place. That distinction is accepted in full. The public release of this dataset does not include an exact admission or discharge date, and the only basis for treating encounter\_id as an ordinal chronology proxy is the UCI data dictionary's description of encounter\_id as a sequential identifier, not any direct evidence that encounter IDs increase monotonically with true encounter time within or across the 130 contributing hospitals. This project does not have, and cannot obtain from this public release, evidence that would confirm or rule out non-monotonic behavior, for example if two or more hospitals' encounters were merged into the source system in a way that interleaves their ID sequences imperfectly with true time.

This is now stated as an explicit, load-bearing assumption rather than an established fact: the entire time-aware train and test split, and the patient-level snapshot's choice of each patient's "first eligible encounter," both depend on encounter\_id being at least approximately monotonic in true time. If that assumption is wrong, meaning if encounter\_id ordering is meaningfully scrambled relative to true chronology, two consequences follow. First, the train and test split would no longer be a genuine chronological split, and the project's claim that the model is evaluated the way it would eventually be used in deployment (trained on earlier patients, tested on later ones) would be weaker than stated elsewhere in this report. Second, the patient-level snapshot's selection of each patient's "first" encounter could occasionally select a later encounter instead, which would not bias the utilization features used as predictors (since number\_outpatient, number\_emergency, and number\_inpatient are already lagged relative to whichever encounter is retained) but could shift which specific encounter's outcome is used as the modeling target for a given patient. Given the absence of hospital identifiers or exact dates in the public data, this assumption cannot be directly tested with this dataset alone; it is disclosed here as a structural limitation of the chosen data source, consistent with Section 5.4, rather than resolved.

## 4.2 Missingness Mechanism Claims: Softened to Match the Available Evidence

The Preprocessing report stated that no field in this dataset was judged to have missingness that was purely mechanical or random. That claim was overly strong. Concluding that missingness is genuinely non-random (as opposed to Missing Completely at Random, or MCAR) is a formal statistical claim that would require dedicated testing, for example Little's MCAR test or a comparison of observed-data distributions between missing and non-missing subgroups, and no such test was run. What this project can defensibly claim, and does claim going forward, is narrower: for each of the nine incomplete fields, there is a plausible, domain-grounded reason to believe the missingness is not MCAR (a physician's decision not to order a lab test, an administrative gap in specialty or payer recording, an unrecorded demographic field), and that reason is sufficient to justify not using a default, distribution-blind imputation strategy such as mean, median, or mode substitution for those fields. This is an assumption grounded in how the data dictionary describes each field's collection process, not a demonstrated statistical fact, and the preprocessing decisions in Section 2.2 and Section 4.2 are sound under that assumption without requiring the stronger, unsupported claim originally made.

## 4.3 Split-Ratio Interpretation: Testing the Train/Test Distribution Gap

The Preprocessing report observed a 2.8-percentage-point gap in readmission rate between the training partition (9.64%) and the held-out test partition (6.82%), and initially characterized the test evaluation as "conservative," implying the gap works in the model's favor. Feedback correctly reframed this: the key issue is that training and test appear to be drawn from a different outcome distribution, and the more important question is why, since that determines whether the gap is a benign artifact of how the snapshot was built or a sign of an evaluation problem.

Three explanations were considered, and they are not mutually exclusive. The first is that the patient-level snapshot design (retaining only each patient's earliest eligible encounter) systematically concentrates the training partition with first-time, earlier-chronology patients and the test partition with a different mix of first-time patients who happen to fall later in the encounter\_id sequence, which is not equivalent to a uniform population shift. Evidence for this explanation comes from the encounter-level benchmark split preserved in Phase 2 (not used for production, but retained specifically for this kind of comparison): applying the identical chronological cut point to the full, non-deduplicated encounter-level data produces a much smaller train/test gap (11.23% vs. 11.68%, a 0.45-point difference) than the patient-level snapshot's 2.8-point gap. Because the same cut point produces a small gap on one version of the data and a large gap on the deduplicated version, the snapshot construction, not the chronological cut point itself, is the more likely primary driver. The second explanation is a genuine secular trend, meaning real change in patient mix, diagnosis coding practice, or hospital discharge behavior across the dataset's 1999 to 2008 span; this cannot be ruled out and would be consistent with the well-documented evolution of diabetes management and inpatient care over that period. The third explanation, addressed directly in Section 4.1, is that the gap is partly an artifact of

encounter\_id not being a perfectly reliable chronology proxy in the first place, in which case the "training" and "test" partitions are not as cleanly separated in true time as their labels suggest.

This project's evidence favors the first explanation as primary, but does not have the tools within the current pipeline to rule out the second or third with confidence. A concrete test that would help distinguish these explanations, and is proposed here as future work rather than completed in this report, is to compute readmission rate and key utilization-feature means within the training partition alone, split further into chronological quartiles or deciles; a smooth trend across those sub-partitions would support the secular-drift explanation, while a rate that is flat within training and only diverges at the train/test boundary would further support the snapshot-construction explanation already favored. Practically, this means the 6.82% test-set base rate, not the pooled 8.94% snapshot-wide figure, remains the correct reference point for interpreting the test-set precision and recall reported in Section 3.3, and the model's decision threshold was appropriately tuned on training-set out-of-fold predictions rather than assumed transferable unchanged from a different prevalence.

#### **4.4 Unsupervised Evidence: Quantitative Support for Excluding PCA and K-Means**

Feedback on the Preprocessing report noted that PCA and clustering were appropriately evaluated rather than merely mentioned, and that excluding them after finding them unhelpful is a reasonable choice, but that the report did not show the evidence behind that choice (variance results or a scree plot for PCA; clustering variables, candidate k values, and a quantitative diagnostic such as silhouette score for k-means). That evidence is now included in full in Section 2.3: the PCA scree plot and component-to-target correlation table (Section 2.3.1, Figures 1 through 3), and the k-means elbow and silhouette sweep across  $k = 2$  through 8 alongside the resulting cluster profile table (Section 2.3.2, Figure 4, Table 3). Both techniques are excluded from the production pipeline on the strength of this evidence, not on interpretability grounds alone.

#### **4.5 Patient Independence: A Residual Clustering Risk That Cannot Be Fully Ruled Out**

Feedback on the Baseline report identified a nuanced but important distinction: confirming zero patient overlap between train and test protects against patient-level leakage and substantially reduces one obvious source of dependence, but it does not, by itself, demonstrate full statistical independence among all observations. That distinction is accepted, and the claim is softened accordingly throughout this report. The zero-overlap check and the one-row-per-patient snapshot design are correct and necessary steps, and they resolve the specific, known non-independence problem this project could observe and measure directly, namely a single patient's repeated encounters appearing in both training and evaluation data. What they do not, and cannot, rule out is dependence introduced by structure this project has no visibility into: this public dataset does not identify which of the 130 contributing hospitals, which physician, or which care team a given

encounter came from, so any hospital-level, provider-level, or regional clustering in discharge practice, coding conventions, or readmission policy is entirely unobservable and therefore uncorrected for. If patients are not evenly and independently distributed across those unmeasured groups relative to the chronological train/test cut (for example, if certain hospitals contributed disproportionately to the earlier or later portion of the encounter\_id sequence), the effective number of independent observations in this analysis is smaller than the row counts reported throughout this document, and reported performance metrics would be somewhat optimistic relative to a hospital that was not represented, or was underrepresented, in training. This is disclosed here as an irreducible limitation of the chosen public dataset rather than a defect in the analysis: the conclusions in this report hold under the assumption that no substantial hospital-level or other unmeasured clustering confounds the chronological split, and that assumption cannot be verified with the fields available in this data.

#### **4.6 Framing the Model as a Screening Tool, Not a Precision Instrument**

Feedback on the Hyperparameter Tuning report suggested making explicit, for a non-technical stakeholder audience, that because absolute PR-AUC remains modest (0.108 to 0.160 across every model family and phase of this project, against a positive base rate of roughly 7 to 10%), the tuned model should be understood as a broad screening or risk-stratification tool rather than a precise, individually reliable predictor of any single patient's readmission. That framing is adopted explicitly here and carried through Section 5: at the selected operating threshold, 92% of patients flagged as high risk will not, in fact, be readmitted within 30 days. The model's appropriate use is to prioritize a larger pool of patients for lower-cost interventions such as a follow-up phone call or a medication review, not to make a high-confidence, individualized clinical determination about any single patient. This framing is restated in Section 5.1 as part of this report's answer to the original research question.

#### **4.7 A1Cresult and A1Cresult\_tested: Redundancy, Its Effect on Feature Importance, and a Recommendation for Future Work**

Feedback on the Hyperparameter Tuning report raised two related points about the two top-ranked features. First, on interpretation: A1Cresult and A1Cresult\_tested ranking as the model's strongest predictors could be misread by a stakeholder as suggesting that ordering the test causes readmission risk, when the more defensible clinical reading is the reverse and more mundane one, that physicians already order A1C tests routinely to monitor insulin dosing and medication regimens in patients whose diabetes looks harder to control, so an abnormal or tested A1C is a marker of a patient a physician already judged to be higher-acuity, not a causal driver of readmission on its own. This report makes that distinction explicit: A1C testing status and result should be read as predictive signal correlated with clinical severity as judged by the treating physician at the time, not as a causal mechanism, and no claim of causation is made anywhere in this report about any feature, consistent with the correlational, observational nature of this entire analysis.

Second, on methodology: A1Cresult and A1Cresult\_tested are very likely strongly dependent on each other by construction, since A1Cresult can only take a non-"not tested" value when A1Cresult\_tested is true. With correlated or redundant predictors, permutation importance can split or distort the importance attributed to each one, because shuffling one feature in isolation creates feature combinations (for example, a shuffled A1Cresult value paired with the true A1Cresult\_tested flag) that do not occur naturally in the data, which can inflate, deflate, or arbitrarily divide the apparent importance between the two correlated features relative to what a model trained without the redundancy would show. This project did not have time within the current phase to refit the tuned model with one of the two features removed and recompute permutation importance, which is the direct test this concern calls for. That refit and comparison is therefore recorded here as a specific, scoped recommendation for the next iteration of this project rather than left as an unaddressed gap: retrain the selected logistic regression specification with A1Cresult\_tested dropped (retaining only the three-level A1Cresult category, which already encodes "not tested" as one of its levels) and recompute test-set permutation importance, to check whether number\_inpatient's rank and relative importance shift once the redundancy is removed. Section 5.3 restates this as a numbered next step.

## 5. Discussion & Next Steps

### 5.1 Conclusions and Key Takeaways

This project set out to answer two questions: whether a patient's 30-day readmission risk can be predicted from routinely captured clinical encounter data, and whether that prediction can be delivered through a live system rather than a one-time report. Both are answered affirmatively, with important qualifications stated directly rather than left implicit.

On the predictive question, the tuned logistic regression model outperforms the naive majority-class baseline by a meaningful margin on the metrics this project committed to in advance: ROC-AUC of 0.608 clears chance-level performance (0.50), and, more importantly for this use case, recall for the readmitted class rose from an unusable 0.25% at the untuned baseline's default threshold to 78.68% once the model was properly tuned and the threshold was selected deliberately. The research question is therefore answered as posed in the original proposal: the trained classifier outperforms the naive baseline on AUC-ROC and recall for the readmitted class by a wide margin, and it is served through a live prediction endpoint, demonstrating both predictive value and deployability. At the same time, absolute PR-AUC remained modest and essentially flat across every model family and every phase of this project (0.108 to 0.160), which is the evidence behind the screening-tool framing adopted in Section 4.6: this model identifies a broad, higher-risk population worth prioritizing for lower-cost outreach, and it should not be read as a precise, individually reliable predictor for any single patient.

On the hypothesis, that patients with more complex or intensive care histories are more likely to be readmitted, the evidence is genuine but partial, and the qualification matters as much as the confirmation. Utilization history is confirmed as an important predictor: `number_inpatient` ranks third in permutation importance, and the K-means Cluster 3 profile independently identifies a small, high-prior-utilization patient segment with a readmission rate roughly double the training-set average, agreement between a supervised and an unsupervised analysis that neither technique was built to guarantee. However, the hypothesis as originally written also predicted utilization-related features would rank among the model's most important predictors overall, and that ranking is led instead by A1C testing status and result, which the original hypothesis did not name. The hypothesis is best described as partially supported and revised by the evidence: care complexity, as measured jointly by prior utilization and by whether and how a patient's diabetes was being actively monitored through A1C testing at this encounter, predicts 30-day readmission risk, with the latter carrying more weight than the former once feature importance is measured consistently. This revision is consistent with, not contradictory to, the underlying reasoning in the original hypothesis: an A1C test ordered at this encounter is itself a marker that a physician judged this patient's diabetes management to be a concern, which is a different but related expression of the same underlying construct, disease severity and care complexity, that motivated the original prediction.

## 5.2 Contextualizing This Work in the Broader Field

This project's PR-AUC and ROC-AUC figures sit at the modest end of what has been reported for readmission prediction on this same dataset and similar diabetic inpatient populations in the published literature; Strack et al. (2014), working with the source dataset this project draws on, reported that HbA1c testing status was independently associated with readmission risk after adjustment, a finding this project's permutation-importance result (A1Cresult and A1Cresult\_tested as the top two predictors) independently corroborates using a different modeling approach and a different, later-collected split of the same underlying data. That convergence across two independent analyses, roughly a decade apart, using different methods, on the specific role of A1C testing is a meaningfully stronger form of evidence than either result alone. Where this project differs from most prior academic treatments of this dataset is not in raw predictive performance but in the surrounding system: an experiment-tracked, registry-managed, API-served model is a different and, for a hospital operations audience, more directly useful contribution than an additional percentage point of ROC-AUC would be. Framed this way, this project's primary contribution to the broader field is less a claim about a new state-of-the-art classifier and more a worked demonstration that a modest, honestly-reported, imbalance-aware classifier can be operationalized end to end on a public clinical dataset without specialized infrastructure, which is a gap the original proposal identified in most prior academic treatments of this dataset.

## 5.3 Recommendations and Future Directions

### *For hospital administration and care management stakeholders*

- Treat the model's output as a prioritization signal for a broader outreach program (follow-up calls, medication review, discharge-planning attention), not as an individualized clinical determination; the low precision at the recommended threshold (8%) means the great majority of flagged patients will not, in fact, be readmitted.
- Confirm the F2-weighted operating threshold (0.43) against actual outreach capacity before deployment; if a smaller, more targeted list is operationally preferable, the F1-maximizing threshold (0.51: 53% recall, 14% precision) is a readily available, more conservative alternative already computed in Section 3.3.
- Consider the K-means Cluster 3 definition (frequent prior utilizers, Section 2.3.2) as a complementary, plain-language targeting input for care coordination staff, used alongside individual model scores rather than as a replacement for them.
- Recognize that any deployment decision made on this model's output should account for the dataset's age (1999-2008) and its single-condition, single-country population; a pilot period against contemporary outcomes is recommended before any operational reliance on the model's specific probability estimates, as opposed to its general prioritization logic.

***For future technical work on this project***

- Retrain the selected model with A1Cresult\_tested removed and recompare permutation importance to the version reported in Section 3.4, directly testing the redundancy concern raised in Section 4.7.
- Split the training partition into chronological sub-groups (quartiles or deciles by encounter\_order) and compare readmission rate and key utilization-feature means across them, as a concrete test of the secular-drift versus snapshot-construction question raised in Section 4.3.
- Complete the Phase 2 stretch goal: wrap the existing training script in an Airflow DAG with a retrain task, an evaluation task that compares a new model's metrics against the current Production model, and a conditional promotion task that only flips the Production tag if the new model wins. The FastAPI service already reads whichever model is tagged Production, so no change to the serving layer is required for this extension.
- Revisit the k-means  $k = 4$  selection with a formal robustness check against  $k = 2$  and  $k = 3$ , given that silhouette score is actually highest at those smaller values (Section 2.3.2), and document whether the Cluster 3 high-risk profile persists at those alternative values of  $k$ .
- Explore the full-longitudinal, patient-grouped encounter-level split retained as a stretch-goal benchmark in Phase 2, to quantify how much predictive performance and feature-importance stability change when repeated patient encounters are treated as independent rows rather than collapsed to one snapshot per patient.

**5.4 Caveats, Limitations, and Sources of Bias**

This section consolidates, in one place, the limitations that are distributed by topic throughout the rest of this report, since a single, explicit accounting is easier for a stakeholder to evaluate than the same information scattered across methods and results sections.

The dataset itself is the largest source of external validity risk. It was collected between 1999 and 2008, predating the HRRP program this project's motivation rests on, predating widespread electronic health record adoption, and reflecting an era of ICD-9 diagnosis coding and diabetes management practice that has materially changed since. Any relationship this model has learned describes historical patterns in this specific population and should be read as a proof of concept for the modeling and deployment pipeline, not as a claim about how a contemporary hospital's patients would behave; a model intended for real deployment would need retraining on contemporary, ideally hospital-specific, data before use. The dataset is also modest in scale (101,766 encounters, reduced to 70,439 independent patients, with an 8.94% positive class further split into training and test) relative to modern, nationally representative alternatives such as the AHRQ Nationwide Readmissions Database, and it covers a single condition population (diabetic inpatients) across a fixed set of 130 hospitals whose identities and case mix cannot be

examined, since neither hospital identifiers nor exact calendar dates are exposed in the public release.

That last point compounds into a specific, unresolved statistical assumption discussed at length in Section 4.5: the zero-patient-overlap check between training and test partitions rules out one concrete form of leakage but cannot rule out clustering by hospital, provider, or region, none of which this dataset makes observable. A second unresolved assumption, discussed in Section 4.1, is that `encounter_id` increases monotonically with true encounter time; this assumption underlies both the chronological train/test split and the patient-level snapshot's choice of each patient's earliest encounter, and it is supported only by how the source data dictionary describes the field, not by direct evidence. If either assumption is materially violated, meaning if there is meaningful unmeasured hospital-level clustering, or if `encounter_id` ordering does not track true chronology closely, this project's evaluation would be more optimistic than a genuinely prospective, hospital-external evaluation would show, and the reported test-set metrics in Section 3.3 should be read with that possibility in mind rather than as a guaranteed floor on real-world performance.

A third limitation concerns interpretation rather than validity: this is an observational, correlational analysis, and no claim of causation is made or should be inferred anywhere in this report, a point made explicit in Section 4.6 specifically because the model's top feature (A1C testing status) is the one most likely to be misread as causal by a non-technical audience. Relatedly, the two top-ranked features (A1Cresult and A1Cresult\_tested) are likely strongly dependent by construction, which can distort how permutation importance divides credit between them, a concern flagged in Section 4.7 and left as a recommended, not completed, follow-up analysis due to time constraints in this phase. Finally, this project made a series of preprocessing and modeling choices, documented and justified throughout Sections 2 and 4, each of which is defensible given the evidence available but none of which is beyond question: the decision to encode missingness as an explicit category rather than impute it, the decision to restrict the modeling dataset to one row per patient, the decision to weight recall over precision through the F2 threshold, and the decision to exclude PCA and k-means from the deployed pipeline. Each of these is a simplifying assumption that, if a stakeholder's priorities or a future analyst's evidence differ from the ones applied here, could reasonably be revisited, and each is written up with enough detail in this report and the underlying repository for exactly that kind of revision to be made.

## 6. Code Availability

All acquisition, cleaning, feature-engineering, modeling, and serving code for this project is version-controlled and publicly available at

<https://github.com/AngelineSetiawan/AngelineSetiawan-hospital-readmission-risk-predictor>.

The repository is organized as five sequential, numbered notebooks corresponding to each phase of this project (data acquisition and EDA; preprocessing and feature engineering; baseline modeling; hyperparameter tuning, evaluation, and unsupervised analysis; MLflow registration and FastAPI serving), plus a serve/ directory containing the production FastAPI application. Raw data (diabetic\_data.csv and IDS\_mapping.csv) is included alongside the processed train and test splits so that every table and figure in this report can be reproduced directly from the repository without re-downloading from UCI.

For reproducibility: all randomness in the modeling notebooks (train/validation splits inside cross-validation, model initialization, and k-means initialization) is seeded from a single constant, `RANDOM_STATE = 42`, defined once at the top of each notebook. Key package versions confirmed directly from executed notebook output include MLflow 3.15.1 and XGBoost 3.2.0; the complete, pinned dependency list for the Python environment used to produce every result in this report is maintained in the repository's environment file rather than restated here, so that it stays current as the environment evolves. The MLflow tracking database (mlflow.db) and the registered model artifact for run `e0c0cf5aedf9490487d9b67f84b4b5ac` are included in the repository so that the exact production model discussed in Section 3.6 can be reloaded and re-served without retraining.

## 7. References

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## Appendix

### A. Data Dictionary and Codebook

The table below lists every variable in the model-ready table used for the final tuned model, consistent with the format used in every prior phase report in this project.

| Variable                       | Data Type    | Source / Status     | Description                                                                                                                                    |
|--------------------------------|--------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| patient_nbr                    | Integer (ID) | Source              | Unique patient identifier; deduplication and grouping key for the patient-level snapshot.                                                      |
| encounter_id                   | Integer (ID) | Source              | Unique identifier for a single inpatient encounter; underlies the ordinal chronology proxy (see Section 4.1 for the assumption this rests on). |
| encounter_order                | Integer      | Engineered          | Sequential rank derived from encounter_id, used as a stand-in for calendar time in the split.                                                  |
| readmitted_30d                 | Binary (0/1) | Engineered / TARGET | Collapsed from the native three-class label; 1 if readmitted within 30 days.                                                                   |
| age                            | Categorical  | Source              | Ten-year age bracket.                                                                                                                          |
| race                           | Categorical  | Standardized        | Includes an explicit missing category for unrecorded values.                                                                                   |
| gender                         | Categorical  | Source              | Patient gender classification.                                                                                                                 |
| admission_type_id_label        | Categorical  | Mapped / clean      | Descriptive admission type; hidden-unknown placeholders consolidated.                                                                          |
| discharge_disposition_id_label | Categorical  | Mapped / clean      | Descriptive discharge destination; excluded from the modeling feature matrix (post-discharge information).                                     |
| admission_source_id_label      | Categorical  | Mapped / clean      | Descriptive referral source; hidden-unknown placeholders consolidated.                                                                         |
| time_in_hospital               | Integer      | Source              | Length of stay in days.                                                                                                                        |
| num_lab_procedures             | Integer      | Source              | Number of lab tests performed during the encounter.                                                                                            |
| num_procedures                 | Integer      | Source              | Number of non-lab procedures performed during the encounter.                                                                                   |
| num_medications                | Integer      | Source              | Number of distinct medications administered during the encounter.                                                                              |
| number_outpatient              | Integer      | Source              | Outpatient visits in the year prior to the encounter (already lagged).                                                                         |
| number_emergency               | Integer      | Source              | Emergency visits in the year prior to the encounter (already lagged).                                                                          |
| number_inpatient               | Integer      | Source              | Inpatient admissions in the year prior to the encounter (already lagged).                                                                      |

| Variable                                         | Data Type          | Source / Status | Description                                                                                                     |
|--------------------------------------------------|--------------------|-----------------|-----------------------------------------------------------------------------------------------------------------|
| diag_1_group /<br>diag_2_group /<br>diag_3_group | Categorical        | Engineered      | Primary, secondary, and tertiary diagnoses grouped into 9 clinical categories.                                  |
| number_diagnoses                                 | Integer            | Source          | Total number of diagnoses recorded for the encounter.                                                           |
| max_glu_serum_tested                             | Binary categorical | Engineered      | Whether a glucose serum test was ordered (Tested / Not Tested).                                                 |
| max_glu_serum                                    | Categorical        | Source          | Glucose serum test result, if performed; 'not tested' retained as its own category.                             |
| A1Cresult_tested                                 | Binary categorical | Engineered      | Whether an HbA1c test was ordered (Tested / Not Tested).                                                        |
| A1Cresult                                        | Categorical        | Source          | HbA1c test result, if performed; 'not tested' retained as its own category. Top-ranked predictor (Section 3.4). |
| change                                           | Categorical        | Source          | Whether diabetic medication dosage was altered during the visit.                                                |
| diabetesMed                                      | Categorical        | Source          | Whether any diabetic medication was prescribed.                                                                 |
| medical_specialty                                | Categorical        | Standardized    | Attending specialty, with missing values represented as an explicit category.                                   |
| payer_code                                       | Categorical        | Standardized    | Payer category, with missing values represented as an explicit category.                                        |

**Table A1. Data dictionary for the model-ready table used in final modeling (49 features before one-hot encoding).**

## B. Additional Summary Table: Full Project Metric History

Table B1 consolidates every headline test-set metric reported across this project's phases into a single table, to make the trajectory from baseline to final model visible in one place.

| Phase         | Model                              | Threshold       | Test PR-AUC | Test ROC-AUC | Test Recall (readmitted) |
|---------------|------------------------------------|-----------------|-------------|--------------|--------------------------|
| Baseline      | Logistic Regression (default)      | 0.50            | 0.1083      | 0.6046       | 0.0025                   |
| Tuned         | Logistic Regression (L1, balanced) | 0.50 (default)  | 0.1086      | 0.6080       | 0.5412                   |
| Tuned (final) | Logistic Regression (L1, balanced) | 0.43 (selected) | 0.1086      | 0.6080       | 0.7868                   |

**Table B1. Test-set metric history across project phases, all evaluated on the same held-out 17,610-patient test set.**